Summary Report

Introduction
This project focuses on data preprocessing and feature engineering for customer purchase analysis. The objective was to clean, transform, and prepare the dataset for machine learning.

Data Issues Observed
The raw dataset contained several challenges:

- Missing values in transaction-related fields
- Skewed numerical features such as income and amount
- Presence of outliers affecting data distribution
- Mixed data types (e.g., product_id containing both text and numbers)
- Categorical variables requiring encoding

Techniques Applied

1. Data Cleaning and Missing Value Handling
Missing values were handled using:

- Mean and median imputation for numerical variables
- Mode imputation for categorical variables
- Advanced methods such as KNN Imputer and MICE algorithm

2. Exploratory Data Analysis (EDA)

- Univariate, bivariate, and multivariate analysis were performed
- Visualizations such as histograms, boxplots, and heatmaps were used

3. Outlier Detection and Handling

- Z-score method
- IQR method
- Percentile method
- Winsorization to cap extreme values

4. Feature Engineering

- Created new features such as purchase_per_day
- Extracted date features like year and month
- Calculated days_since_last_purchase

5. Encoding Techniques

- Label Encoding for binary variables
- One Hot Encoding for categorical variables
- Ordinal Encoding for ordered categories

6. Feature Scaling

- StandardScaler, MinMaxScaler, and RobustScaler were applied
- ColumnTransformer was used for applying multiple transformations

7. Transformations

- Log, square root, and reciprocal transformations
- PowerTransformer (Yeo-Johnson method) for normalization

8. Binning and Binarization

- Income was grouped using quantile binning
- Created binary feature for frequent buyers

Observations

- Handling missing values improved data completeness
- Outlier treatment stabilized the dataset
- Feature engineering enhanced predictive capability
- Scaling ensured balanced feature contribution

Conclusion
The dataset was successfully cleaned, transformed, and prepared for machine learning. The preprocessing pipeline improved data quality and ensures better predictive performance.